

What's Picking Up

For solar	Some key players	What's latest
Cells and modules	Tata BP, Moser Baer Photo Voltaic, Reliance Solar, Titan Energy, Indo Solar, Kotak Urja, Premier Solar Systems, SunTechnics, Maharishi Solar, Central Electronics, Lanco Solar, Bergen Group India, XL Telecom & Energy, Surana Ventures, Steca Bergen, Kyocera Solar, Solar Semiconductor, Applied Materials, Signet Solar, etc	The latest in the cell and module market are the crystalline silicon cells and the thin-film of amorphous silicon. The crystalline cells have higher efficiency as compared to amorphous. Power modules 9 Range of 210 to 300 Wp used in rooftop systems and free field installations
Batteries	Su-kam, Numeric, Microtek, Regnant, Amara Raja, Exide, HPL, etc	Sealed maintenance-free batteries, without water, according to Amit Kumar Yadav, AGM, Su-Kam
Glass	Borosil	Borosil is setting up a plant in India to start production in January 2010. Latest are light transmission >91.5 per cent; standard thickness of glass: 3.2 mm and 4 mm
Chemicals and consumables	Linde Gas, BOC, Horiba, Dow Corning, etc	Shift towards using fluorine chemical instead of NF3 gas to clean chemical vapour deposition (CVD) chambers, according to Vivek Bhatnagar, Sales Manager, BOC
Test & Measurement	National Instruments, Forevision Metrology, Agilent Technologies, etc	Solar cell and module testers, sun simulators, spectral response measurement units, sun sensors, UV chambers, climatic chambers, simulation software, according to PEN Savant, Application Engineer, Forevision Metrology
Cables & connectors	LAPP India	Inter-module cables, used for connecting one solar module to another; Output cables, used for connecting solar PV system to electric junction box; MC output cables, each cable includes MC connector, non-metallic cord connector with 10AWG type 'USE' 600V rated outdoor black wire
Thin films	Moser Baer, NexPower, etc	Crystalline technology
Solar products	Gautam Polymers, Scone solar systems, EPL India	Highest revenue generator are solar water heater and solar lantern

already announced FITs at 500 MW. A total of 3275 MW was filed for this FIT benefit, to be made available by March 2014. Indian government should study the example set by Spain whose FITs policy of 2007-2008 drove the global PV sales through the roof. Its PV sales went from 600 MW in 2007 to 2511 MW in 2008.

Ways to support domestic market growth

For India to emerge as a large market, it is imperative to have a centralised single window clearance for solar

projects. This will make a big difference in the time it takes to connect to the grids, says Arya. An extension of the proposed FIT to more than the current 10 years, and higher rates, is a good first step, he says. "The FIT does not currently promise a convincing ROI. That will make the risks too high for banks to provide financing. And project financing is nowadays the most crucial issue in getting projects moving," says Grover.

Another possible solution to develop the Indian PV market is to set up or stimulate new solar funds to

guarantee project financing. Initial investments are too high in almost all Indian PV projects, and banks are hesitant nowadays. On the other hand, players have realised that returns are good and secured, and are willing to pay higher interest rates as well. So, a win-win scenario seems possible with market development and good financial returns in exchange.

Many Indian manufacturers are now preferring to go in for good financing and not rely on government subsidy. "Subsidies will take years to come, and when they come they will be beneficial, but at the moment we are going ahead without subsidies," says Grover.

The high cost of installation also come as an obstacle to the growth of the solar industry. According to Dr Gay, the cost of installation is likely to go down further as volumes and scale go up. Nandedkar also agrees, "Today, installation per megawatt is approximately Rs 170 million, but in years to come it will be reduced as the volume and technology improvement will bring down the cost."

Challenges and hindrances

The serious fall in global demand for PV cells and modules, and the competition from more than 400 cell and module manufacturers worldwide, has created a situation of oversupply, which will hit the solar industry hard. Exports to Europe have also declined due to worsened dollar-euro ratio. All this poses a serious threat to the export of Indian solar cells and modules.

Another challenge faced by the country is to bring down the cost of manufacturing solar PVs. Finances and the limited technical knowhow remain the key barriers for the solar projects in India. "Making people aware of the business prospects of solar and convincing them to invest in it is quite challenging," points out Grover. But Arya is hopeful while he concludes, "Our expectation is that within the next two quarters things should improve enough that it makes sense to go ahead with our plans." ●